

VIKRAMADITYA SINGH

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Profile

Senior Software Engineer with 5+ years of experience designing and building event-driven backend systems, microservices, and cloud-native platforms. Strong background in Java, OCI, AWS, databases, distributed systems, and performance optimisation, with experience owning services end-to-end from architecture and API design through deployment and observability. Experienced in Spec-Driven Development (SDD), with a track record of mentoring engineers and establishing reusable engineering foundations across teams. Effective in cross-team collaboration, translating complex technical concepts into clear solutions, and driving solutions amid evolving product and engineering requirements.

Experience

ORACLE

Bangalore, India

Senior Member of Technical Staff

April 2025 – present

- Designed and developed an event-driven microservice that ingests clinical-data changes and generates safety-critical advisories.
- Designed and implemented a low-latency notification microservice enabling automatic data refresh for the Patient List UI.
- Created a reusable microservice scaffolding template and shared integration framework for the platform's clinical-data layer, decreasing new-service setup time by 30%.
- Implemented an asynchronous PDF generation microservice for patient-list reporting using Thymeleaf and OpenHTMLToPDF.
- Owned CI/CD pipelines for the Patient Advisory, Patient List, Care Notification, and Consult microservices using OCI CI/CD, and built Grafana dashboards and alerts for service monitoring and proactive issue detection.
- Designed and implemented the core Patient List and Patient Detail APIs, then enhanced their performance by 60% through query optimization and caching.
- Converted multiple OpenSearch queries to SQL for the Oracle ATP database, simplifying and optimizing them to improve performance by 12%.
- Developed a data migration module for Acute Organiser to migrate multiple clinical concepts from Gen 1 to Gen 2 EHRC.
- Implemented 5+ FHIR clinical concepts into data models and onboarded them in the EHRC framework, enabling the platform to support new clinical data types and use cases.
- Designed and implemented an MCP service and reusable framework using FastMCP to query Acute Organizer services.
- Applied a Spec-Driven Development (SDD) approach to design and implement consult microservice, reducing development time by 40%.

CISCO

Bangalore, India

Software Engineer III

August 2021 – April 2025

- Architected and scaled the backend for the Policy Analyser and Optimiser (PAO) platform from the ground up, including anomaly detection, automated remediation, and a multi-version policy converter, enabling analysis across 300+ cloud firewall configurations.
- Built a pub/sub framework on AWS SQS/SNS that delivered real-time analysis updates across 4+ downstream services.
- Reduced FMC device listing and management page load time by 96% through pagination, caching, query optimization.
- Cut Intrusion Prevention System (IPS) and Network Analysis Policy (NAP) snapshot creation time by 70%, accelerating policy deployment from management center to firewalls for the enterprise customers.
- Led backend development for a Zero Trust network access policy feature on Firewall Management Center, spanning APIs, persistence, validation, auditing, and telemetry, including automated certificate auto-enrollment that reduced device onboarding time by 25%.
- Extended FMC's API surface to support Snort3 rule querying against the MITRE framework, and shipped audit-log/delta-preview visibility for access-policy rule changes, improving traceability for security teams.
- Delivered end-to-end support for Elephant Flow Detection and Threat Detection, spanning configuration, deployment, PDF reporting, and audit telemetry.
- Owned CI/CD pipeline automation for build, test, and deployment, and built custom Datadog dashboards with proactive alerting, reducing incident detection time by 70%.

SECURONIX

Pune, India

SDE Intern

August 2020 – April 2021

- Developed email microservice and automated incident report generation using Spring Framework and Jasper Reports.

- Developed a gRPC detector for Snort 2.9.16 and integrated it with FTD to enable alert generation.
- Contributed to the preprocessor framework for scanning the message content transferred via gRPC for malicious payloads.

Publication

Inception Time Model for Structural Damage Detection Using Vibration Measurements

March 2024

In book: Fourth Congress on Intelligent Systems (pp.103-122), Springer, Singapore, 31 March 2024

DOI: 10.1007/978-981-99-9040-5_7

Technical Skills

Languages: Java, SQL, JavaScript, Python, C

Backend & Frameworks: Spring Boot, Spring Framework, Hibernate, Micronaut, React.js, Next.js, REST APIs, Microservices, Event-Driven Architecture

Databases & Caching: PostgreSQL, Oracle ATP, MongoDB, Redis, Lucene, OpenSearch

Cloud & Infrastructure: OCI, AWS (SQS, SNS, EC2), Kafka, OCI Queue, Docker, Terraform, Kubernetes, OCI CI/CD

Observability & Tools: FastMCP, Datadog, Grafana, JMeter, VisualVM, Maven

Projects

ExpenseSplit | *React.js, NestJS, PostgreSQL, Redis, Drizzle ORM, Docker*

demo

- Built a modular expense-sharing platform with workspace-based multi-tenancy, Google OAuth/JWT authentication, CI/CD automation, and OpenAPI-driven API contracts.
- Implemented a debt-simplification algorithm that computes net balances and minimizes settlement transactions.

Meditor | *Neutrino.js, React.js, Local LLMs, SQLite*

code

- Built a Markdown editor with integrated local LLM inference for document summarization.
- Designed a knowledge vault using Retrieval-Augmented Generation (RAG) to retrieve relevant notes and generate context-aware answers to user queries.

Domain Generating Algorithm Detection | *Python, Keras, Pandas, Google Colab*

code

- Detected algorithmically generated domains with 96% accuracy and classified them by DGA variant with 87% accuracy.
- Designed a two-stage approach combining feature extraction, clustering, and classification to identify known DGA families and detect previously unseen variants.

NeuralCook | *NextJS, NestJS, PostgreSQL, GitHub Actions*

demo | code

- Built and deployed a full-stack technical publishing platform using NextJS and NestJS with PostgreSQL persistence.
- Automated build and deployment workflows using GitHub Actions.

Bookmarker | *React.js, NestJS, Redis, PostgreSQL*

demo

- Built a bookmark-sharing platform with a NestJS backend and PostgreSQL persistence, supporting categorization, auto-tagging, and search.
- Integrated Redis caching to improve frequently accessed bookmark retrieval and reduce database load.

Achievements

Smart India Hackathon

2019

Finalist for hardware edition

- Built a Flutter/Firebase irrigation assistant and deployed a LoRaWAN network to extend connectivity and overcome range limitations.

Awards

- Cisco Recognition Awards: Connect Everything (4×), Going Above and Beyond (2×), Living Cisco's Principles (1×), Driving the Highest Level of Achievement (1×)

Education

Indian Institute of Information Technology, Guwahati

2017 – 2021

Bachelor of Technology in Computer Science Engineering

Cum. GPA: 8.48